

Payroll Regeneration API Guide

1. Regenerate Payroll

Endpoint

POST /api/v1/payroll/{payrollId}/regenerate

Path Parameter

Parameter	Type	Description
payrollId	Long	ID of the currently active payroll version

Authentication

Bearer Token is required.

Allowed Roles

- SUPER_ADMIN
- HR_ADMIN
- PAYROLL_ADMIN

2. Request Body

The request body is **optional**.

Only send the fields that need to be changed.

If a field is not sent, the backend keeps the value from the existing payroll.

Request Body

```
{
  "totalWorkingDays": 30,
  "workedDays": 28,
  "lopDays": 2,

  "basicSalary": 35000.00,
  "hra": 10000.00,
  "specialAllowance": 0.00,
  "medicalAllowance": 0.00,
  "travelAllowance": 0.00,
  "bonus": 5000.00,
  "otherAllowance": 0.00,

  "pf": 1800.00,
  "esi": 0.00,
  "professionalTax": 0.00,
  "incomeTax": 0.00,
  "otherDeduction": 0.00,

  "remarks": "Salary corrected before approval"
}
```

All fields are optional.

3. Request Fields

Attendance

Field	Type	Required
<code>totalWorkingDays</code>	Integer	No
<code>workedDays</code>	Integer	No
<code>lopDays</code>	Integer	No

Earnings

Field	Type	Required
basicSalary	Decimal	No
hra	Decimal	No
specialAllowance	Decimal	No
medicalAllowance	Decimal	No
travelAllowance	Decimal	No
bonus	Decimal	No
otherAllowance	Decimal	No

Deductions

Field	Type	Required
pf	Decimal	No
esi	Decimal	No
professionalTax	Decimal	No
incomeTax	Decimal	No
otherDeduction	Decimal	No

Other

Field	Type	Required	Validation
remarks	String	No	Maximum 500 characters

4. Important Request Body Rules

Send Only Changed Fields

For example, if only Basic Salary and Bonus are changed:

```
{
  "basicSalary": 35000.00,
  "bonus": 5000.00
}
```

Do **not** send all other unchanged fields.

Omitted Fields

If a field is omitted, the old value is retained.

Example:

```
{
  "basicSalary": 35000.00
}
```

Only `basicSalary` changes.

All other values come from the existing payroll.

Null Values

`null` does **not** clear the existing value.

Example:

```
{
  "basicSalary": null
}
```

The existing `basicSalary` will be retained.

5. Fields That Must NOT Be Sent

The frontend must not send the following fields during regeneration.

Calculated Fields

- grossSalary
- lopAmount
- totalDeduction
- netPayable
- payableDays

Payroll Identity / Version

- id
- payrollNumber
- version
- active
- status

Employee / Payroll Snapshot

- employee
- salaryStructure
- payrollMonth
- employeeName
- employeeCode
- dateOfJoining
- departmentName
- designationName
- panNumber
- uanNumber
- bankName
- accountNumber
- ifscCode

Approval / Payment

- approvedBy
- approvedDate
- paymentDate
- paymentReference

These fields are managed by the backend.

6. Request Examples

Example 1 – Change Salary

```
POST /api/v1/payroll/500/regenerate
Content-Type: application/json
```

```
{
  "basicSalary": 35000.00,
  "bonus": 5000.00
}
```

Example 2 – Change Attendance

```
POST /api/v1/payroll/500/regenerate
Content-Type: application/json
```

```
{
  "lopDays": 1,
  "remarks": "LOP corrected"
}
```

Example 3 – Change Multiple Fields

```
{
  "basicSalary": 35000.00,
  "hra": 12000.00,
  "bonus": 5000.00,
  "pf": 1800.00,
  "lopDays": 1
}
```

Example 4 – No Changes

```
POST /api/v1/payroll/500/regenerate
```

No body is required.

This creates a new version using the existing payroll values.

7. Backend Calculations

The backend recalculates all calculated fields.

Gross Salary

```
Gross Salary =  
    basicSalary  
  + hra  
  + specialAllowance  
  + medicalAllowance  
  + travelAllowance  
  + bonus  
  + otherAllowance
```

LOP Amount

```
LOP Amount =  
    (Gross Salary / totalWorkingDays) × lopDays
```

Total Deduction

```
Total Deduction =  
    pf  
  + esi  
  + professionalTax  
  + incomeTax  
  + otherDeduction  
  + lopAmount
```

Net Payable

```
Net Payable =  
    Gross Salary - Total Deduction
```

The frontend should **not** send these calculated values.

8. Response

Success

```
200 OK
```

The response contains the **new payroll version directly**.

There is no additional response wrapper.

Example Response

```
{  
  "id": 501,  
  "payrollNumber": "PR-202608-0006",  
  "version": 2,  
  "active": true,  
  "employeeId": 101,  
  "employeeCode": "EMP000101",  
  "employeeName": "Jitendra Prajapati",  
  "dateOfJoining": "2024-04-01",  
  "departmentName": "Engineering",  
  "designationName": "Software Engineer",  
  "panNumber": "ABCDE1234F",  
  "bankName": "State Bank of India",  
  "accountNumber": "123456789012",  
  "ifscCode": "SBIN0001234",  
  "payrollMonth": "2026-08-01",  
  "status": "GENERATED",  
  
  "totalWorkingDays": 30,  
  "workedDays": 28,
```

```
"lopDays": 2,
"payableDays": 28,

"basicSalary": 35000.00,
"hra": 10000.00,
"specialAllowance": 0.00,
"medicalAllowance": 0.00,
"travelAllowance": 0.00,
"bonus": 5000.00,
"otherAllowance": 0.00,

"grossSalary": 50000.00,
"lopAmount": 3333.33,

"pf": 1800.00,
"esi": 0.00,
"professionalTax": 0.00,
"incomeTax": 0.00,
"otherDeduction": 0.00,

"totalDeduction": 5133.33,
"netPayable": 44866.67,

"approvedBy": null,
"approvedDate": null,
"paymentDate": null,
"paymentReference": null,

"remarks": "Salary corrected before approval",

"createdAt": "2026-09-18T10:15:30",
"updatedAt": "2026-09-18T10:15:30"
}
```

9. Version Behavior

Regeneration creates a **new payroll record**.

Before Regeneration

Field	Existing Payroll
ID	500
Version	1
Active	true
Status	GENERATED

After Regeneration

Old Payroll

Field	Value
ID	500
Version	1
Active	false
Status	SUPERSEDED

New Payroll

Field	Value
ID	501
Version	2
Active	true
Status	GENERATED

The old payroll is **not deleted**.

10. Important Status Behavior

Frontend Should Allow Regeneration For

- DRAFT
- GENERATED
- APPROVED

An **APPROVED** payroll can be regenerated, but the new version will require approval again.

Frontend Should Hide Regeneration For

- PAID
 - SUPERSEDED
 - CANCELLED
-

11. Error Responses

Error Format

```
{
  "success": false,
  "message": "Total Deduction cannot exceed Gross Salary.",
  "errorCode": "BAD_REQUEST",
  "path": "/api/v1/payroll/500/regenerate",
  "timestamp": "2026-09-18T10:15:30"
}
```

Common Errors

HTTP Status	Error Code	Reason
400	BAD_REQUEST	Business validation failed
400	VALIDATION_FAILED	Invalid field value
400	INVALID_REQUEST	Invalid JSON/type
403	ACCESS_DENIED	User does not have permission
404	RESOURCE_NOT_FOUND	Payroll not found
500	INTERNAL_SERVER_ERROR	Server error

Validation Examples

Paid payrolls cannot be regenerated.

Only the active payroll version can be regenerated.

Worked days + LOP days cannot exceed total working days.

Gross Salary must be greater than 0.

Total Deduction cannot exceed Gross Salary.

Net Payable cannot be negative.

12. Frontend API Implementation

Axios – With Request Body

```
const response = await api.post<PayrollResponse>(
  `/api/v1/payroll/${payrollId}/regenerate`,
  {
    basicSalary: 35000,
    bonus: 5000,
    lopDays: 1
  }
);
```

Axios – Without Request Body

```
const response = await api.post<PayrollResponse>(
  `/api/v1/payroll/${payrollId}/regenerate`
);
```

After Success

The regenerated payroll has a **new ID**.

Use:

```
response.data.id
```

Do not continue using the old payroll ID for the newly generated version.

13. TypeScript Request Model

```
export interface RegeneratePayrollRequest {
  totalWorkingDays?: number;
  workedDays?: number;
  lopDays?: number;

  basicSalary?: number;
  hra?: number;
  specialAllowance?: number;
  medicalAllowance?: number;
  travelAllowance?: number;
  bonus?: number;
  otherAllowance?: number;

  pf?: number;
  esi?: number;
  professionalTax?: number;
  incomeTax?: number;
  otherDeduction?: number;

  remarks?: string;
}
```

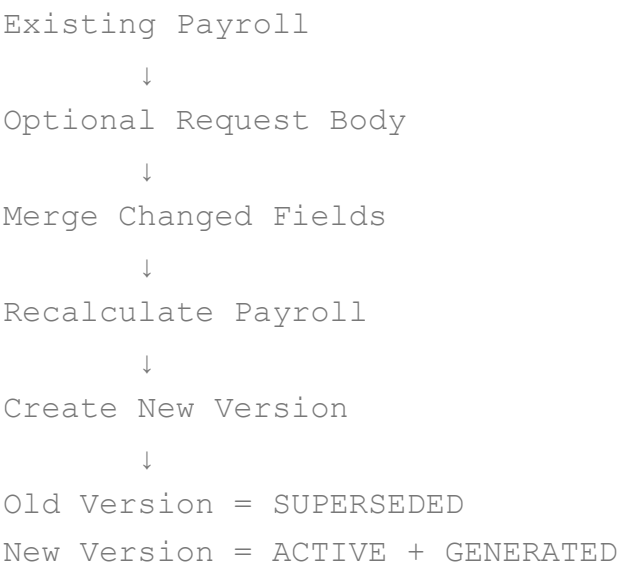
All properties are optional because the API supports partial updates during regeneration.

14. API Summary

Operation	Method	Endpoint
Get Payroll	GET	/api/v1/payroll/{payrollId}

Operation	Method	Endpoint
Regenerate Payroll	POST	/api/v1/payroll/{payrollId}/regenerate

Regeneration Flow



Main Point for Frontend

Send only the fields the user wants to change.

The backend handles:

- Merging unchanged values
- Calculating Gross Salary
- Calculating LOP Amount
- Calculating Total Deduction
- Calculating Net Payable
- Calculating Payable Days
- Payroll versioning
- Activating the new version
- Superseding the old version
- Business validation
- Approval reset for regenerated approved payrolls

The frontend should **not** calculate or send backend-managed fields.